

Answer Key & Teacher Notes

Session 1 – Using Properties to Rewrite Expressions • Session 2 – Expanding and Factoring Expressions

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Using Properties to Rewrite Expressions

TALK PROMPTS

- Why does rewriting an expression with properties never change the value it gives?
- Between your two answers you used a property to move terms around. Which one, and why is it allowed?

PARTNER TASK

Two Steps Together. Partner A distributes in $5(j + 6) + 4j$; Partner B combines like terms and names the property used to reorder. Both check with $j = 1$ (39 and 39). Switch roles with $2(k + 10) + 4k$.

WATCH FOR

- Multiplying only the first term inside the parentheses ($4(q + 5)$ written as $4q + 5$).
- Combining unlike terms ($7m + 8$ written as $15m$).
- Using the Commutative Property for subtraction or division.

SESSION 2 • TEACHER NOTES

Expanding and Factoring Expressions

TALK PROMPTS

- How are expanding and factoring opposite moves? Explain what each one does.
- One answer removed parentheses and the other added them. How can both moves keep the value the same?

PARTNER TASK

Both Forms. Partner A expands $6(y + 7)$; Partner B factors $14b + 35$ with the GCF. Trade papers and check each answer by doing the opposite move.

WATCH FOR

- Factoring out a common factor that is not the greatest ($16y + 40 = 2(8y + 20)$).
- Multiplying only the first term when expanding ($5(w + 8) = 5w + 8$).
- Dividing only one term by the GCF when factoring ($18x + 24 = 6(18x + 4)$).

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.EE.A.3	1	1	2	1	1	2	3	2	3	1	1
S1 • Form B	6.EE.A.3	1	1	2	1	1	2	3	2	3	1	1
S2 • Form A	6.EE.A.3	1	1	2	1	1	2	2	3	3	1	1
S2 • Form B	6.EE.A.3	1	1	2	1	1	2	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

Distributive Property: Multiply the outside number by **each** term inside the parentheses.

Commutative Property: Changing the **order** when adding or multiplying does not change the answer.

Associative Property: Changing the **grouping** when adding or multiplying does not change the answer.

like terms: Terms with the same **variable** part.

- Expressions are **equivalent** when they give the same value for every value of the variable.
- To simplify: distribute, put like terms together, then **combine** them.

Example: Distribute: $4(m + 2) = 4m + 8$. Reorder: $4m + 3m + 8$ (Commutative Property). Combine like terms: $4m + 3m = 7m$. Answer: $4(m + 2) + 3m = 7m + 8$

YOU TRY

1. a. $2x + 10$ b. $5y + 3$
2. $7c + 16$; check: $6(3) + 1 + 4 = 23$ and $7 + 16 = 23$

EXIT TICKET

1. a. $5x + 15$ b. $8n + 8$
2. $7b + 8$
3. **B** (Associative)

PRACTICE FORM A

1. a. $4n + 20$ b. $8x + 6$ c. $15y + 18$ d. $5a + 11$
2. a. Associative b. Commutative c. Identity
3. a. $3n + 12$ b. $8n + 12$ c. 28 and 28
4. $7p + 23$
5. $10q + 19$
6. No. $24 \div 6 = 4$ but $6 \div 24 = 1/4$. The Commutative Property works for adding and multiplying, not dividing.
7. Sample: $3w + 6 + w$ (Distributive); $3w + w + 6$ (Commutative); $4w + 6$ (combine like terms)
8. a. $3a + 18$ and $2a + 8$ b. $5a + 26$ c. 51 tickets
9. The 4 must multiply every term inside, so $4 \times 5 = 20$ too. $4(q + 5) = 4q + 20$.
10. **B** ($5x + 20$)
11. **B** ($7n + 6$)

Self-check word PROPERTY

PRACTICE FORM B

1. a. $6n + 42$ b. $11x + 12$ c. $13y + 10$ d. $5a + 16$
2. a. Associative b. Commutative c. Identity
3. a. $4n + 12$ b. $10n + 12$ c. 42 and 42
4. $9p + 16$
5. $10q + 17$
6. No. $30 \div 5 = 6$ but $5 \div 30 = 1/6$. The Commutative Property works for adding and multiplying, not dividing.
7. Sample: $2w + 10 + 3w$ (Distributive); $2w + 3w + 10$ (Commutative); $5w + 10$ (combine like terms)
8. a. $4a + 12$ and $2a + 10$ b. $6a + 22$ c. 46 tickets
9. The 3 must multiply every term inside, so $3 \times 7 = 21$ too. $3(k + 7) = 3k + 21$.
10. **C** ($7b + 42$)
11. **B** (every term inside)

Self-check word IDENTITY

GUIDED NOTES

expand: Multiply to **remove** the parentheses.

factor an expression: Write it as a **product**: a shared factor times a sum.

term: One **part** of an expression, separated by + or – signs.

greatest common factor: The **biggest** number that divides both numbers evenly.

- Factoring **undoes** expanding, so you can check a factored form by expanding it.
- Factor out the **GCF** so the terms inside share no other common factor.

Example: The GCF of 18 and 27 is **9**. Divide each term: $18x \div 9 = 2x$ and $27 \div 9 = 3$. Check: $9 \times 2x + 9 \times 3 = 18x + 27$. Answer: $18x + 27 = 9(2x + 3)$

YOU TRY

- $8x + 12$ b. $5(3y + 4)$
- $8(2n + 3)$; check: $8 \times 2n + 8 \times 3 = 16n + 24$

EXIT TICKET

- $7a + 35$ b. $6(2b + 3)$
- $10(2m + 3)$
- B** $(9y + 18)$

PRACTICE FORM A

- $8c + 24$ b. $10d + 20$ c. $6y + 42$ d. $7n + 14$
- $4(3m + 5)$ b. $5(3k + 7)$ c. $7(2b + 5)$ d. $6(3x + 4)$
- $4t + 36$ b. 5 c. $5(4t + 9)$
- $5x + 30$
- $9z + 24$
- No; 8y and 20 still share 4. GCF form: $8(2y + 5)$
- $6(5c + 7)$; check: $6 \times 5c + 6 \times 7 = 30c + 42$
- $12(2p + 3)$ b. 12 equal rows with $2p + 3$ plants in each row c. $24(4) + 36 = 132$ and $12(11) = 132$
- Lucia forgot to multiply 8 by 5. The 5 multiplies every term inside: $5(w + 8) = 5w + 40$.
- B** $(5(2x + 3))$
- B** $(6y + 24)$

Self-check word **FACTORS**

PRACTICE FORM B

- $9w + 36$ b. $12d + 15$ c. $5y + 45$ d. $8n + 24$
- $6(4q + 3)$ b. $7(3k + 4)$ c. $5(2b + 5)$ d. $6(2x + 5)$
- $6s + 48$ b. 7 c. $7(4s + 7)$
- $6x + 18$
- $8z + 15$
- No; 6y and 15 still share 3. GCF form: $9(2y + 5)$
- $9(a + 3)$; check: $9 \times a + 9 \times 3 = 9a + 27$
- $6(3p + 5)$ b. 6 equal rows with $3p + 5$ plants in each row c. $18(4) + 30 = 102$ and $6(17) = 102$
- Amina forgot to multiply 3 by 7. The 7 multiplies every term inside: $7(k + 3) = 7k + 21$.
- B** $(4(2n + 3))$
- A** (greatest common factor)

Self-check word **PRODUCT**

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 · USING PROPERTIES TO REWRITE EXPRESSIONS

1. a. $6n + 12$ b. $12x + 3$ c. $10 + 5y$ d. $7a - 21$ e. $6b + 10$ f. $10m - 10$
2. a. $6(30 + 4) = 180 + 24 = 204$ b. $8(50 + 2) = 400 + 16 = 416$
3. a. $9c + 12$ b. $3x + 13$ c. $7m + 10$ d. $5t + 18$
4. $6x + 24$; when $x = 5$: $6 \times 5 + 24 = 54 \text{ ft}^2$ (check: $6 \times 9 = 54$)
5. She did not multiply the 4 by 2. $2(3x + 4) = 6x + 8$, so $6x + 8 + x = 7x + 8$.

SHEET 2 · EXPANDING AND FACTORING EXPRESSIONS

1. a. $15x + 6$ b. $8y + 56$ c. $12 + 8n$
2. a. $3(2x + 5)$ b. $4(5y + 8)$ c. $7(2a + 7)$ d. $9(4 + 3k)$ e. $20(2m + 3)$ f. $11(b + 3)$
3. $8x + 56 = 8(x + 7)$, so the length is $x + 7$ units
4. Sample: $2(6x + 12)$, $3(4x + 8)$, $4(3x + 6)$, or $6(2x + 4)$.
GCF form: $12(x + 2)$.
5. No. $6n$ and 9 still share a factor of 3. The GCF is 15: $15(2n + 3)$.
6. **B** ($8(2x + 5)$)
7. **C** ($18a + 12$)